



Comsats Universty Sahiwal Campus

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Submission Date:

03/10/2025

Subject:

Human Computer Interaction

Assignment # 01

Topic: Implementation of Don Norman Principles on WhatsApp and YouTube

1. Visibility

- **WhatsApp:** Essential options like chats, calls, and status are always visible on the main screen. Message ticks (✓, ✓✓, ✓✓ blue) provide visibility of delivery status.
- **YouTube:** The play/pause, volume, and progress bar are always visible during playback, making controls easy to find.

2. Affordance

- **WhatsApp:** The microphone icon affords recording a voice message when pressed and held. The paperclip affords attaching files.
- **YouTube:** The play button triangle affords playing a video, while the bell icon affords enabling notifications.

3. Visible Constraints

- **WhatsApp:** Users cannot send a message without internet access, which constrains actions to avoid confusion. File sharing is limited to supported formats.
- **YouTube:** Only signed-in users can upload or comment, constraining unauthorized actions. Some features like live streaming are restricted until account verification.

4. Mapping

- **WhatsApp:** Video call and voice call buttons are mapped clearly at the top of each chat. Swipe left/right gestures map to actions like reply or delete.
- **YouTube:** Keyboard shortcuts (space = play/pause, arrow keys = forward/back) map logically to expected behaviours.

5. Feedback

- **WhatsApp:** When a user types, “typing...” appears to the recipient. Sending an image shows a loading circle until it is delivered.
- **YouTube:** When liking a video, the thumbs-up button changes color instantly. A buffering symbol shows network issues.

6. Conceptual Models

- **WhatsApp:** Users see chats as a “conversation thread” similar to SMS, making the app intuitive.
- **YouTube:** The app provides a “virtual TV” conceptual model where users can search, play, pause, and skip videos, just like using a remote.

7. Consistency

- **WhatsApp:** The interface is consistent across iOS, Android, and Web. The same icons and layouts are used in all versions.
- **YouTube:** The video layout (title, description, like/dislike, comments) remains consistent across mobile, desktop, and TV apps.

8. Transfer Effects

- **WhatsApp:** People transfer knowledge from using SMS and phone calls to WhatsApp, so they quickly understand messaging, calling, and contact lists.
- **YouTube:** Watching and controlling videos feels similar to using a DVD player or TV remote, helping users transfer previous experiences.

9. Population Stereotypes

- **WhatsApp:** Uses familiar icons like the phone for calls and camera for pictures, which are widely understood across populations.
- **YouTube:** The red play triangle is universally recognized as a video symbol, making it instantly understandable.

10. Individual Differences

- **WhatsApp:** Provides adjustable font sizes and dark mode for users with visual impairments.
- **YouTube:** Offers captions, playback speed control, and accessibility features to support different user needs.

11. Cultural Association

- **WhatsApp:** Emoticons and stickers often reflect local cultural contexts, making conversations relatable.
- **YouTube:** Content recommendations and trending sections adapt to regional cultures, showing what is popular in a specific country.

Conclusion

Both **WhatsApp** and **YouTube** successfully implement **Don Norman's extended design principles** to enhance interactivity and usability. From **visibility and affordances** to **cultural associations and individual differences**, these applications ensure that users from diverse backgrounds can interact with them naturally and effectively. Their design choices explain why they are two of the most widely adopted digital platforms in the world.